Isolated staged legacy metadata preparation

Source-only implementation receipt, 2026-10-18. This is the explicit adjacent nineteenth candidate EXP-ECDLP-daee4d; membership in the user's historical eighteen contracts remains unauthenticated. No complete original toy panel, native scientific trial, mathematical claim, hypothesis transition, provider action or publication was performed by this implementation task.

Authority: handoff TASK-20261018-72227c and Coordinator decision DEC-20261018-43f3b8 assign tools/legacy_staged_manifest.py, its synthetic test file, and this task directory. The unchanged original specification, trial-plan-v1.json, run.py and check.py are retrieved internal source evidence. No old experiment path or sealed run was written.

The old driver resolves global output paths from its own source location. Its checker resolves the same paths from the working directory. A simple raw/manifest adapter would therefore write old global paths, omit worksheet-note.md, and either encounter write-once files or append to a sealed predecessor. This adapter copies the exact source, specification and old plan into a new isolated workspace for each stage, preserving the same old-relative paths and original child argv. The legacy interpreter must resolve to the actual declared native adapter runtime.

Stage 0 archives ten expected regular files: four source/spec/plan copies, three local old-run outputs, and three global freeze files including worksheet-note.md. Stage 1 builds a separate workspace, verifies the validated native Stage 0 receipt and its complete artifact hash set, validates that predecessor's saved envelope/archive, and copies only the three exact Stage 0 globals. It then archives thirteen potential files: the same four source copies, three imported freeze files, three Stage 1 local outputs and up to three new global outputs. It never adds descendants to Stage 0. The old relative RUN identities remain visible inside each archive; two fresh outer native RUN IDs will be allocated by the Coordinator. These are dependent stages, not two independent arithmetic routes.

Four fixed driver artifacts are proposed per stage: manifest.yaml (new native envelope), raw-result.json (exact child raw bytes), legacy-execution.json (measured child commands/CWD/resources, source/supervisor/predecessor bindings and actual/expected/absent path inventory), and legacy-workspace.tar (every actual staged regular file, retaining the old local manifest and RESULTS.md). The native supervisor supplies its separate eight generated files. All twelve paths per stage require prospective archive/write-scope coverage. The source closure binds this helper plus all four original dependencies; actual runtime and Python/PyYAML versions are recorded without model attestations.

The tar reader rejects noncanonical paths, duplicates, links and nonregular members. The adapter compares the durable archive against actual staged file bytes before removing the scratch workspace. The checker reconstructs only validated regular bytes into a fresh temporary workspace before replaying the unchanged old checker. It does not use tar.extractall. Snapshot failure or host/process loss can leave partial tar/workspace descendants; broad native run-directory archival scope must retain those failures as well. Missing artifacts stay missing. Failed child phases retain their actual partial bytes and a failure receipt where possible, and never emit a successful envelope. Consumed output/workspace paths cannot be retried in place.

The old seed named null_seed is the actual fixed integer 20261007. Stage 0 freezes it without enumeration or RNG. Stage 1 retains random.Random(20261007+p+1009*a+b) exactly. No random seed or independent replication was invented. The old ascending search, N in (1024,4096), two selected ordinary toys, exclusion of a=b=1, Hamming rule, negation handling, controls, excess formula and thresholds remain unchanged. The existing scalar-recovery control selects W_aff itself; the |W|/N base rate and order_E!=p+1 ordinary filter are preserved rather than strengthened by metadata.

Software validation: thirteen synthetic staging/metadata controls passed in 3.273 seconds. They exercise separate Stage 0/1 fixtures, exact bytes/source/globals and sealed-predecessor immutability; native minimum-envelope acceptance; required predecessor receipt, exact native artifact coverage and digest drift; fixed-seed type and source/command/entrypoint tampering; no-overwrite; driver/checker failure retention; honest O-IMPEDIMENT absence; imported-freeze mutation refusal; tar traversal/duplicate/link refusal; receipt timestamps and bool/int substitution. The final logs are checks/synthetic-03.stdout.log and .stderr.log. Synthetic02 passed the original twelve controls; synthetic01 retained a unittest method-name collision in the test harness before any fixture ran. The native validator emitted an existing unclosed-file ResourceWarning;

its schema error list was empty for both synthetic stages.

The old checker validates outcome vocabulary and file presence only. It does not independently verify point counting, the selected group order, ranking/overlap arithmetic, controls or threshold classification. Canonical result.valid records that the preserved instrument and metadata checks passed; certificate.kind is none, with no scientific support/falsification inferred. The prospective layout explicitly carries this limitation. Native schema acceptance, hashes and a fresh envelope do not repair the old arithmetic-verification gap.

Deliverables: trial-layout.json contains non-executable placeholders, the complete prospective legacy_staged block, all original fixed cells/controls/stopping rules and potential paths; legacy-staged-block.json is the exact proposed block; source-closure.sha256 binds five runtime dependencies; implementation-receipt.json and ARTIFACTS.sha256 bind the preparation. The source-only diagram is preparation-flow.dot/.svg/.png and this report has a freshly rendered PDF. No scientific quantities were measured, so the diagram shows staging and unresolved gates instead of a research-results plot. Existing canonical research graphs were not changed: no scientific evidence or frontier was produced.

Next authority remains with the Coordinator: allocate a new successor EXP and two primary RUNs, freeze the actual specification/plan/approval/runtime and full closure, publish native claims/handoff/queue/archive coverage, then use the existing run admission gates. This preparation does not itself admit or authorize scientific execution. Independent arithmetic verification remains an explicit missing obligation for interpreting any future legacy result.

